The rank-one phase-retrieval incidence variety has exactly 2^N irreducible components

Candidate counterexample packet for arXiv:1506.00674

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Status. Candidate counterexample, likely valid, pending human review. The source's incidence variety is not irreducible in general: when all projection ranks are one it has exactly 2^N irreducible components.

1 Source question and scope

Edidin studies phase retrieval by N orthogonal projections on an M-dimensional real vector space and introduces a complex incidence variety $\mathcal{I}_{k_1,\ldots,k_N,M}$ in the proof of generic phase retrieval [1]. Proposition 4.3 controls the union of the irreducible components containing the real locus. Remark 4.4 then states that the full incidence variety is not asserted to have the computed dimension because its irreducibility is not known. The source passage is on PDF page 7:

$\Phi = (P_1, \ldots, P_N)$.

We will show that when $N \geq 2M - 1$ the variety $\mathcal{I}$ contains an open set of complex dimension less than that of $\mathcal{P}_{k_1} \times \ldots \times \mathcal{P}_{k_N}$ that contains all of the real points of $\mathcal{I}$. This means that $(\mathcal{I})_{\mathbb{R}}$ has real dimension less than $\sum_{i=1}^{M} k_i(M-k_i)$. Hence for generic projections $P_1, \ldots, P_N$ there are no non-zero real vectors x, y such that $\langle P_i x, y\rangle = 0$ for all i. In other words $\mathcal{A}_\Phi$ is injective for generic collections of projections $P_1, \ldots, P_N$ with $N \geq 2M-1$.

Proposition 4.3. *There is an open subset of $\mathcal{I}$ which contains $\mathcal{I}_{\mathbb{R}}$ and has dimension $\sum_{i=1}^{N} k_i(M-k_i) + 2M - 2 - N$. In particular if $N \geq 2M-1$ this open set has dimension strictly smaller than* $\dim \prod_{i=1}^{N} \mathcal{P}_{k_i}$.

Remark 4.4. Note that since we do not know that $\mathcal{I}$ is irreducible we are not asserting that $\mathcal{I}$ has dimension $\sum_{i=1}^{N} k_i(M-k_i) + 2M - 2 - N$. Instead, we are proving that the union of the irreducible components of $\mathcal{I}$ that contain all of the real points has this dimension.

Figure 1: Proposition 4.3 and Remark 4.4 of arXiv:1506.00674, PDF page 7.

The projective factors displayed as $\mathbb{P}^{N-1}$ in the source formula are a typographical slip: the surrounding text has $x, y \in \mathbb{C}^M \setminus \{0\}$, so the factors used below are $\mathbb{P}^{M-1}$.

This packet gives a negative answer to the irreducibility uncertainty and an exact component classification for the all-rank-one case. It does not alter the generic phase-retrieval theorem.

2 Definitions

Put $V = \mathbb{C}^M$, $M \geq 2$, and let $b(v,w) = v^\mathsf{T} w$ be the standard nondegenerate symmetric bilinear form. For $1 \leq k \leq M-1$, let

$$\mathcal{P}_k(M) = \{P \in \mathbb{C}^{M\times M} : P^\mathsf{T} = P,\ P^2 = P,\ \operatorname{tr} P = k\}.$$

For ranks $k_1, \ldots, k_N$, define the incidence variety

$$\mathcal{I}_{k_1,\ldots,k_N,M} = \{(P_1,\ldots,P_N,[x],[y]) : b(y, P_i x) = 0 \quad (1 \leq i \leq N)\}$$

inside $\prod_i \mathcal{P}_{k_i}(M) \times \mathbb{P}(V) \times \mathbb{P}(V)$.

Let

$$U = \{[u] \in \mathbb{P}(V) : b(u,u) \neq 0\}.$$

Every $[u] \in U$ determines

$$P_u = \frac{uu^\mathsf{T}}{u^\mathsf{T} u} \in \mathcal{P}_1(M).$$

3 Proof intuition

For a rank-one projection, the incidence equation splits:

$$b(y, P_u x) = \frac{b(u,y)b(u,x)}{b(u,u)} = 0.$$

Thus each measured line must be orthogonal either to x or to y. There are 2^N ways to make those choices. Each choice gives a closed irreducible piece: after fixing the choice, the relevant lines form an open part of a product of projective bundles over $[x]$ and $[y]$. Finally, $x = e_1$, $y = e_2$, and image lines alternating between $\mathbb{C}e_2$ and $\mathbb{C}e_1$ give a point belonging to exactly one chosen piece. Hence the pieces are distinct maximal irreducible subsets, not merely a redundant closed cover.

4 Exact component theorem

Lemma 1 (Rank-one parametrization)**.** *The map*

$$U \longrightarrow \mathcal{P}_1(M), \qquad [u] \longmapsto P_u,$$

is an isomorphism of varieties. Under this isomorphism,

$$b(y, P_u x) = 0 \quad \Longleftrightarrow \quad b(u,x) = 0 \ \text{ or } \ b(u,y) = 0.$$

Proof. The displayed formula for P_u is unchanged when u is rescaled, is symmetric and idempotent, and has image $\mathbb{C}u$. Conversely, if $P \in \mathcal{P}_1(M)$, symmetry gives $\ker P = (\operatorname{im} P)^\perp$. Idempotence gives $V = \operatorname{im} P \oplus \ker P$, so the image line is nondegenerate. Writing it as $\mathbb{C}u$ therefore gives $b(u,u) \neq 0$ and forces $P = P_u$. The construction and its inverse (taking the image line) are regular on these varieties. Direct multiplication gives

$$b(y, P_u x) = \frac{b(u,y)b(u,x)}{b(u,u)},$$

which proves the final equivalence. □

For $a \geq 0$, define

$$Z_a = \{([x],[u_1],\ldots,[u_a]) \in \mathbb{P}(V) \times U^a : b(x,u_j) = 0 \ (1 \leq j \leq a)\},$$

with $Z_0 = \mathbb{P}(V)$.

Lemma 2. *Every Z_a is irreducible and*

$$\dim Z_a = M - 1 + a(M-2).$$

Proof. Over $\mathbb{P}(V)$, the hyperplanes $x^\perp$ form the fibers of a rank $M-1$ vector bundle K: it is the kernel of the everywhere-surjective bundle map

$$V \otimes \mathcal{O}_{\mathbb{P}(V)} \longrightarrow \mathcal{O}_{\mathbb{P}(V)}(1), \qquad u \longmapsto b(x,u).$$

The a-fold fiber product of $\mathbb{P}(K)$ over $\mathbb{P}(V)$ is locally a product of the irreducible base with $(\mathbb{P}^{M-2})^a$; hence it is irreducible and has dimension $M-1+a(M-2)$. The condition that every u_j be nonisotropic removes a closed subset. The resulting open subset is Z_a, and it is nonempty (take $x = e_1$ and all $u_j = e_2$). Therefore Z_a is irreducible of the stated dimension. $\square$

Theorem 3 (All-rank-one component classification)**.** *For $M \geq 2$ and $N \geq 1$, the variety $\mathcal{I}_{1,\ldots,1,M}$ has exactly 2^N irreducible components. Every component has dimension*

$$2M - 2 + N(M-2) = N(M-1) + 2M - 2 - N.$$

Moreover, every component contains real points.

Proof. Use Lemma 1 to replace each P_i by a nonisotropic image line $[u_i]$. For a word $\varepsilon = (\varepsilon_1,\ldots,\varepsilon_N) \in \{X,Y\}^N$, let C_ε be the closed subvariety defined by

$$\begin{cases} b(u_i,x) = 0, & \varepsilon_i = X, \\ b(u_i,y) = 0, & \varepsilon_i = Y. \end{cases}$$

The conditions are equivalently $P_i x = 0$ and $P_i y = 0$, respectively, so they are closed in the original projection coordinates. Lemma 1 gives the finite closed cover

$$\mathcal{I}_{1,\ldots,1,M} = \bigcup_{\varepsilon \in \{X,Y\}^N} C_\varepsilon.$$

If a entries of ε equal X, then, after reordering the u_i coordinates,

$$C_\varepsilon \cong Z_a \times Z_{N-a}.$$

Lemma 2 shows that C_ε is irreducible and has dimension

$$[M-1+a(M-2)] + [M-1+(N-a)(M-2)] = 2M-2+N(M-2).$$

It remains to show that these irreducible pieces are exactly the components. Take real vectors $x = e_1$ and $y = e_2$. For each i, choose

$$u_i = \begin{cases} e_2, & \varepsilon_i = X, \\ e_1, & \varepsilon_i = Y. \end{cases}$$

Then $b(u_i,u_i) = 1$. If $\varepsilon_i = X$, the line $\mathbb{C}u_i$ is orthogonal to x but not to y; if $\varepsilon_i = Y$, it is orthogonal to y but not to x. The resulting real point belongs to C_ε and to no C_δ with $\delta \neq \varepsilon$. Thus no C_ε is contained in another one. A finite closed cover by irreducible subsets with this noncontainment property is precisely the irreducible-component decomposition. The witnesses also prove the last assertion. $\square$

The dimension in Theorem 3 is exactly the formula in source Proposition 4.3, $\sum_i k_i(M - k_i) + 2M - 2 - N$, specialized to $k_i = 1$. Thus all 2^N components containing real points have the same expected dimension.

Corollary 4 (Any rank-one factor forces reducibility)**.** *Suppose exactly $r \geq 1$ of $k_1, \ldots, k_N$ equal one. Then $\mathcal{I}_{k_1,\ldots,k_N,M}$ has at least 2^r irreducible components. In particular, it is reducible.*

Proof. Applying Lemma 1 to the r rank-one indices gives a cover by 2^r closed pattern subsets. Each pattern has a point belonging to no other pattern: take $x = e_1$, $y = e_2$, choose the rank-one image lines as in the proof of Theorem 3, and for every remaining index choose a coordinate orthogonal projection of rank k_i. Such a projection is diagonal, so $e_2^\top P_i e_1 = 0$.

Choose an irreducible component through each pattern-exclusive point. An irreducible subset of a finite union of closed sets lies in one of them. Hence components chosen from two different exclusive points cannot coincide, giving at least 2^r components. $\square$

5 Verification and limitations

The factorization and pattern witnesses were checked independently using exact rational arithmetic in dimensions 2 through 6 and for all patterns through $N = 4$; the reusable checker is included in the packet. This is only a sanity check. The proof of irreducibility of each C_ε is the projective-bundle argument in Lemma 2.

The result answers the source's general irreducibility uncertainty negatively and gives an exact classification when all ranks are one. It does not classify the case $2 \leq k_i \leq M - 1$ for every i. It also does not expose a gap in the source: Remark 4.4 carefully avoids using irreducibility, and Proposition 4.3 remains compatible with the component calculation above.

6 Novelty check and review recommendation

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:1506.00674 and close phase-retrieval incidence terms. No exact prior result was present. Bounded web/arXiv searches used the exact remark phrase, the title and "incidence variety irreducible/reducible," and close variants involving orthogonal projections and components. They found the source and later phase-retrieval literature but no paper giving this 2^N component classification. This is evidence of apparent novelty, not an exhaustive citation review.

Recommendation: send to a human algebraic geometer or phase-retrieval expert. The principal review points are the variety isomorphism in Lemma 1 and the mixed-rank component-count argument in Corollary 4.

References

[1] D. Edidin, *Projections and phase retrieval*, arXiv:1506.00674 (2015), especially Proposition 4.3 and Remark 4.4.

[2] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977, Chapter I.